

Credit Card Fraud Detection Using Cost-Sensitive Hybrid Sampling and Gradient Boosting Case Study for a Digital Banking Environment

1. Background and Business Context Digital payments have grown rapidly over the last decade, with a substantial share of retail transactions now processed through credit and debit cards. Along with this growth, financial institutions face increasing exposure to fraudulent transactions, chargebacks, and regulatory scrutiny. Fraud patterns are dynamic: attackers continuously adapt their behavior in response to new security controls, and simple rule-based systems can quickly become ineffective or too restrictive for genuine customers. A mid-sized digital bank (referred to here as the Bank) operates in multiple regions and processes hundreds of thousands of card transactions per day. Historically, the Bank relied on a combination of manual reviews and rule-based triggers, such as fixed limits per region or merchant category. While these rules captured some fraud, they generated a high volume of false positives, overwhelming the fraud operations team and frustrating genuine customers whose transactions were unnecessarily declined. The Bank initiated a data science project with the objective of building a machine learning based fraud detection system that can detect more fraudulent transactions while keeping false positives under tight control. The solution must be explainable enough for risk and operations teams and suitable for near real-time deployment in production systems.

2. Problem Statement and Objectives The core problem is to predict whether a given credit card transaction is fraudulent or legitimate at the time it is initiated. The challenge is that fraud is extremely rare compared to legitimate activity, leading to a highly imbalanced classification problem. Traditional metrics such as overall accuracy are misleading because a model that always predicts legitimate can still achieve above 99% accuracy but provide zero business value. The business objectives for the solution are as follows: Improve recall for fraudulent transactions, so that a higher percentage of actual frauds are detected before authorization. Maintain precision at a level that is acceptable for customer experience and operational load, so the number of unnecessary alerts and manual reviews remains manageable. Provide a clear way to adjust the operating point (decision threshold) to reflect changing risk appetite or special campaigns (for example, festive seasons, new card launches). Deliver a model that can be integrated into the Bank s existing microservice architecture and called via a standard REST API. Translating these objectives into technical terms, the model should maximize metrics that focus on the minority (fraud) class, such as the F2-score (which emphasizes recall), precision recall AUC (PR-AUC), and cost-sensitive measures that incorporate the different business cost of false negatives (missed fraud) and false positives (incorrectly blocked legitimate transactions).

3. Data Description and Preprocessing For this project, anonymized card transaction data is used, similar in structure to widely used public benchmark datasets. Each row represents a single card transaction with a set of numerical features derived from original raw fields through secure transformations such as PCA. Key fields include: Time: Time elapsed (in seconds) since the first transaction in the dataset. V1 V28: Transformed numerical features capturing cardholder behavior, merchant characteristics, and transaction patterns. transaction. Class: Target label (0 = legitimate, 1 = fraudulent). Amount: Monetary value of the The dataset exhibits severe class imbalance: fraudulent transactions represent well below 1% of the total volume. Standard data preprocessing steps include: Handling missing values and basic data quality checks. Standardizing numerical features using z-score scaling to stabilize model training. Splitting the data into training, validation, and test sets with stratification to preserve class ratios. Carefully applying resampling techniques only on the training set, to avoid information leakage into validation and test sets.

4. Methodology and Model Architecture

4.1 Handling Imbalanced Data Because of the extreme imbalance, the project uses a cost-sensitive hybrid sampling strategy rather than simple undersampling: Random undersampling reduces the size of the majority class to a more manageable ratio while keeping diverse legitimate patterns. SMOTE (Synthetic Minority Over-sampling Technique) then generates synthetic fraud samples to balance the classes in the training set. This hybrid strategy aims to give the model enough examples of fraud behavior without discarding too much information from legitimate transactions.

4.2 Model Choice: CSH-XGB The main predictive engine is an ensemble of gradient boosted decision trees, referred to as CSH-XGB (Cost Sensitive Hybrid Sampling with Gradient Boosting). The model uses: A robust tree-based algorithm (such as XGBoost or an equivalent gradient boosting implementation). Class weights that emphasize the fraud class during training. Hyperparameters tuned for stability and generalization, including depth, number of estimators, learning rate, and sampling fractions. Gradient boosted trees are well suited to this task because they can capture non-linear relationships, interactions between features, and are relatively resilient to monotonic transformations of the inputs.

4.3 Threshold Optimization and Evaluation Strategy After training the model with hybrid sampling and class weights, predicted probabilities are obtained for the validation set. Instead of using a fixed threshold of 0.5, a range of thresholds is evaluated to compute precision, recall, and the F2-score. The threshold that maximizes the F2-score is chosen as the operating point, as it gives more weight to recall while remaining mindful of precision. Evaluation uses stratified train/validation splits to preserve the original class distribution. Core metrics include: ROC-AUC: General discriminatory power. PR-AUC (Average Precision): Quality of ranking with a F1 and F2 scores: Balanced and recall-focused metrics for fraud focus on the positive (fraud) class. detection. Confusion matrices: To quantify the absolute number of true positives, false positives, true negatives, and false negatives.

5. Model Deployment as a FastAPI Microservice To integrate the machine learning model into the Bank's technical ecosystem, a lightweight RESTful microservice is implemented using FastAPI. The service exposes three main endpoints: GET /health Returns model load status and basic health information to facilitate monitoring. POST /predict Accepts a single transaction payload with features Time, V1 V28, and Amount, and returns a fraud probability, the decision threshold used, and a binary decision (fraud vs. legitimate). POST /predict/batch Accepts a batch of transactions in a single call and returns predictions for each transaction. The FastAPI application loads a serialized model object that contains both the pre-processing pipeline (scaling and resampling configuration), the trained gradient boosting model, and the optimal decision threshold. The service is stateless, making it easy to deploy behind an API gateway, scale horizontally with multiple instances, and integrate with existing authorization and logging frameworks. In a production environment, the Bank would containerize this service using Docker and deploy it on a Kubernetes cluster or an equivalent orchestration platform. Standard practices such as blue green deployments, health checks, autoscaling, and centralized logging are used to ensure reliability and observability.

6. Business Impact and Illustrative Results In a pilot evaluation using historical data, the cost-sensitive hybrid model delivered strong performance compared to the rule-based system and simple baseline models: Fraud recall increased significantly, meaning that a larger share of fraudulent transactions was detected before authorization. Precision for fraud alerts remained within operational limits, so the fraud operations team did not experience an unmanageable increase in workload. Overall PR-AUC and F2-score showed that the new system is more effective at prioritizing high-risk transactions than previous approaches. From a business perspective, even a small absolute increase in detected frauds can translate into substantial monetary savings for the Bank, especially when combined with reduced reputational risk and regulatory confidence. At the same time, careful threshold selection ensures that declines or challenges to legitimate customers remain within acceptable boundaries, protecting customer satisfaction and retention. The ability to adjust the decision threshold also allows the Bank to react quickly to emerging threats or seasonal risk spikes. For example, during major shopping festivals, the Bank may temporarily favor higher recall (slightly more false positives) to mitigate increased fraud attempts, then revert to a more balanced operating point after the peak period.

7. Operationalization and Monitoring Deploying a fraud detection model is not a one-time exercise. The Bank needs a robust MLOps process to monitor model performance and retrain when necessary. Key elements include: Data drift monitoring to detect changes in transaction patterns, feature distributions, or customer behavior. Performance tracking on live data, using delayed labels from confirmed fraud/legitimate cases. Automated retraining pipelines that periodically refresh the model with the latest data, subject to validation and approval. Model versioning and rollback mechanisms to ensure that any newly deployed model can be reverted quickly if unintended behavior is observed. In addition, explainability and transparency are crucial. While gradient boosted trees are more complex than simple rules, feature importance scores and partial dependence plots can help

analysts and risk managers understand which factors drive model decisions. This supports internal validation, regulator discussions, and continuous improvement of the risk strategy.

8. Limitations and Future Enhancements Despite its strong performance, the current solution has important limitations: It treats transactions mostly as independent events, even though cardholder behavior evolves over time. Sequence models (for example, LSTM or transformer-based architectures) could be explored in future work to capture temporal patterns. The hybrid sampling strategy is applied offline; real-time systems must be carefully designed so that models do not see unrealistic data distributions when deployed. The model uses anonymized, transformed features that may limit direct interpretability. Additional engineered features (such as velocity measures, geolocation patterns, and merchant-level risk) could further enhance performance. Future enhancements may include: Combining the gradient boosting model with a rule engine so that obvious, well-understood fraud scenarios are still captured by simple deterministic rules, while the model focuses on subtle patterns. Using active learning, where high-uncertainty cases are prioritized for manual review and later fed back into model training. Exploring federated learning approaches that allow collaborative fraud detection models across institutions without sharing raw sensitive data. Overall, the case study demonstrates that a cost-sensitive hybrid sampling strategy combined with gradient boosting and a modern FastAPI deployment approach can deliver a practical, high-impact fraud detection solution that balances accuracy, customer experience, and operational feasibility for a digital bank.